ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION **WINTER SEMESTER, 2023-2024**

DURATION: 1 HOUR 30 MINUTES **FULL MARKS: 75**

CSE 4711: Artificial Intelligence

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses. The symbols have their usual meanings.

1. In Figure 1, a simplified map shows 6 areas connected via directional roads, each associated with a positive integer travel time. A robot plans to travel from area S to area G using a search algorithm to find an optimal path. The robot will break ties in alphabetic order when choosing a node from the fringe.

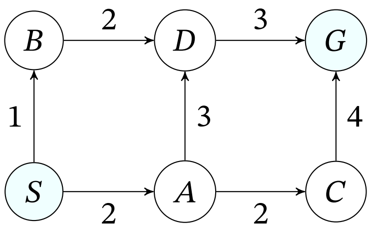


Figure 1: A simplified map of a city for Question 1.

a) Recommend a search algorithm for the robot with a brief justification. 3 (CO3) (PO3)

Solution:

Uniform Cost Search (UCS).
Given no heuristic is available to the robot, UCS will find the optimal path.

Rubric:

- 1 point for the recommendation.
- 2 points for the justification.

b) Determine the optimal path using the algorithm you recommended. Also, show the order in which the states are expanded. $2+2$ (CO1) (PO1)

Solution:

Optimal path: $S \rightarrow B \rightarrow D \rightarrow G$
States expanded: $S \rightarrow B \rightarrow A \rightarrow D \rightarrow C$
(or, if tree-search variant is used) States expanded: $S \rightarrow B \rightarrow A \rightarrow D \rightarrow C \rightarrow D$

Rubric:

- 2 points for the optimal path
- 2 points for the expansion order

c) Suppose you are designing a heuristic function h for the given scenario. You have come up with all the values except $h(B)$ as shown in Table 1. 3×5 (CO1) (PO1)
For each of the following scenarios, write the possible range of values for $h(B)$ with a brief explanation:

i. h will be admissible.

Table 1: Incomplete heuristic function for Question 1.c

Area	S	A	B	C	D	G
h	6	4	?	2	2	0

Solution:

To make h admissible, $h(B)$ has to be less than or equal to the optimal cost of going B to G, which is 5. And the minimum possible value for a heuristic is trivial heuristic, which makes it 0.
The range is: $0 \leq h(B) \leq 5$.
Rubric:

- 1 point for each correct range.
- 3 points for the explanation.

ii. h will be consistent.

Solution:

The consistency conditions that involve the state B are:

- $h(B) \leq c(B, D) + h(D)$
  $\rightarrow h(B) \leq 2 + 2$
  $\rightarrow h(B) \leq 4$
- $h(S) \leq c(S, B) + h(B)$
  $\rightarrow 6 \leq 1 + h(B)$
  $\rightarrow 5 \leq h(B)$

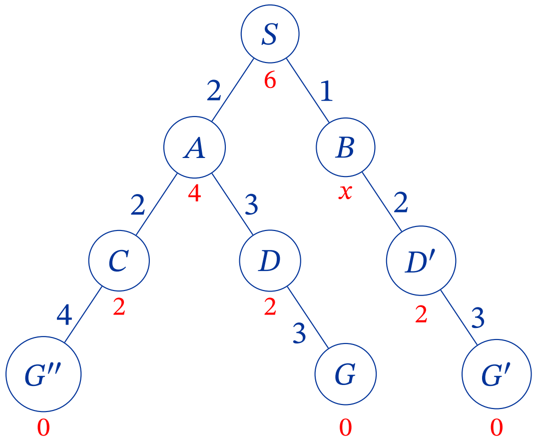
There is no intersection between the two ranges. So it is not possible.
Rubric:

- 2 points for the decision.
- 3 points for the explanation.

iii. Using h, perform A* tree search to return the optimal path.

Solution:

Let $h(B) = x$.
The search tree depicts the edge cost beside each edge [Blue colored] and the heuristic cost below each node [Red colored].



The f-value for each node is as follows:

- $f(S) = 0 + 6 = 6$
- $f(A) = 2 + 4 = 6$
- $f(B) = 1 + x$
- $f(C) = 4 + 2 = 6$
- $f(D) = 5 + 2 = 7$
- $f(D') = 3 + 2 = 5$
- $f(G'') = 8 + 0 = 8$
- $f(G) = 8 + 0 = 8$
- $f(G') = 6 + 0 = 6$

To ensure that A* tree search finds the optimal solution, we need to pull out G' from the fringe before G or G''. $f(S)$, $f(D')$, and $f(G')$ are already less than that of $f(G)$ and $f(G'')$. As a result, if $f(S)$, $f(D')$, or $f(G')$ get into the fringe, they will come out before the suboptimal goal nodes. We need to ensure that $f(B) \leq f(G) = f(G'')$ (equal works due to alphabetic ordering) to get $f(B)$ out of the fringe before the suboptimal goal nodes. We get:

$$f(B) \leq f(G) = f(G'')$$
$$1 + x \leq 8$$
$$x \leq 7$$

Since we are not aiming for admissibility or consistency, the lower range can be any value less than or equal to 7.
The range is: $-\infty \leq h(B) \leq 7$.
Rubric:

- 1 point for each correct range.
- 3 points for the explanation.

2. a) Assume that to find a path from S to G for the problem shown in Question 1, you have defined a set of states, s; a set of actions, A including a $NoOp$ action that has no effect; a successor function, $Successor(s, a)$; a start state, S; and a goal test, $isState == G$?. 8 (CO2) (PO2)

Unfortunately, the robot does not know any search algorithm! All it has is a Constraint Satisfaction Problem (CSP) solver. Reformulate the problem as a CSP to find a path.

Solution:

By analyzing the graph in Figure 1 (or, using the search tree in the solution of Question 1.c)iii), we can find that the maximum number of steps required to determine the plan is 4.

We can define variables S_0, S_1, S_2, and S_3 for the state at each time step, with the domain of each being the set of states s; and variables A_0, A_1, A_2, and A_3 for the actions to be taken at each time step, with the domain of each being A.
The constraints are:

- $S_0 = S$;
- $(S_0 = G) \wedge (S_1 = G) \wedge (S_2 = G) \wedge (S_3 = G)$;
- $S_{t+1} = Successor(S_t, A_t)$, where $0 \leq t \leq 2$.

This CSP will always find a solution if there is one. The plan will be given by the action variables.
Rubric:

- 1 point for identifying the number of steps required
- 2 points for the variables
- 2 points for the domains
- 3 points for the constraints

b) "When presented with a choice, we should choose the variable that is least constrained but the value that is most constraining in a CSP search." - Do you agree with the statement? Justify your decision. $1 + 4$ (CO1) (PO1)

Solution:

No.
Choosing the most constrained variable makes sense because it chooses a variable that is (all other things being equal) likely to cause a failure, and it is more efficient to fail as early as possible (thereby pruning large parts of the search space).
The least constraining value heuristic makes sense because it allows the most chances for future assignments to avoid conflict.
Rubric:

- 1 point for the correct decision.
- 2 points for the justification of the most constrained variable
- 2 points for the justification of the least constraining value

c) Sudoku is a logic-based number puzzle where the goal is to fill a 9×9 grid so that each row, column, and 3×3 subgrid contains the digits 1 to 9. The puzzle starts with a partially filled grid and has only one solution when properly designed. 4 (CO3) (PO3)
Evaluate the effectiveness of the Local Search with Min-Conflicts Heuristic in solving Sudoku, compared to the backtracking approach.

Solution:

It is possible to solve Sudoku problems using Local Search. However, it is not as effective as the backtracking search. This is because there are two types of conflicts: a conflict with one of the given numbers and a conflict between two numbers that are placed by solvers. We may design the min-conflict heuristic to recognize the difference between these two situations to resolve the issue.

Rubric:

- 1 point for recognizing that it is possible to solve Sudoku using local search
- 2 points for identifying the problem
- 1 point for providing a possible solution to solve the problem.

3. The game tree for a game between Altair and Basim is shown in Figure 2.

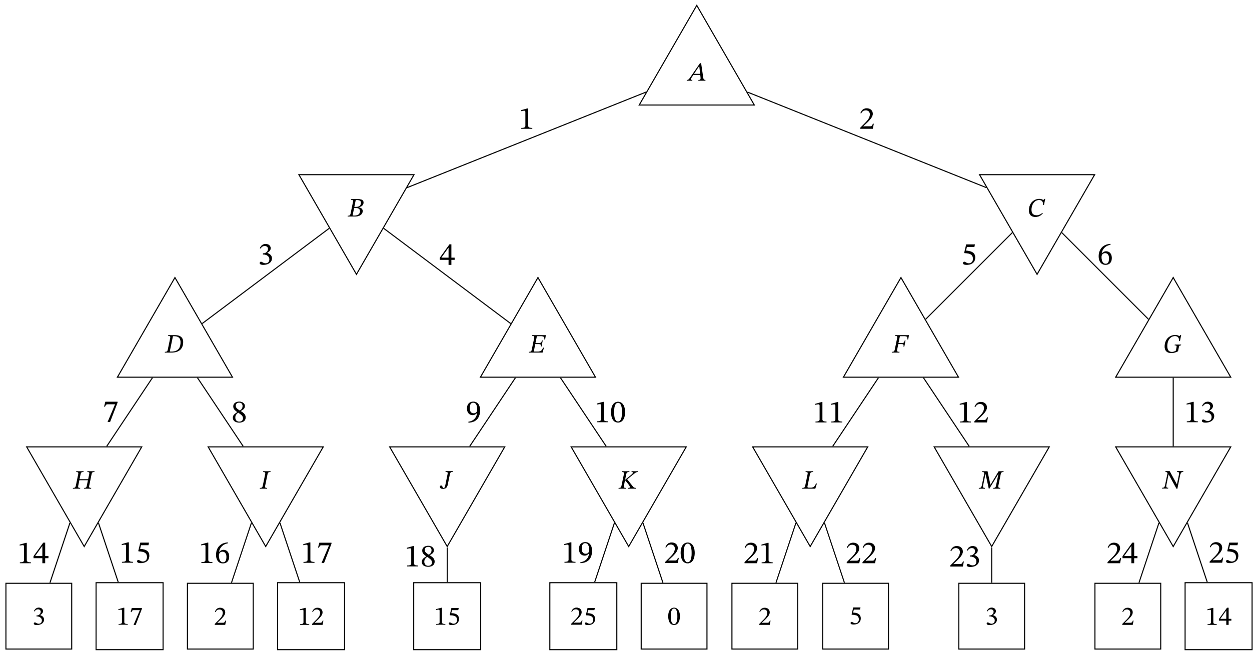


Figure 2: Game tree for Question 3

Here, the non-terminal nodes are labeled with capital letters and branches with numbers. Ties between branches are broken in ascending order. Altair moves first.

a) Recommend the optimal action for Altair showing detailed steps. 8 (CO1) (PO1)

Solution:

- $H = 3$
- $I = 2$
- $J = 15$
- $K = 0$
- $L = 2$
- $M = 3$
- $N = 2$
- $D = \max(3, 2) = 3$
- $E = \max(15, 0) = 15$
- $F = \max(2, 3) = 3$
- $G = 2$
- $B = \min(3, 15) = 3$
- $C = \min(3, 2) = 2$
- $A = \max(3, 2) = 3$

Take the action corresponding to the branch 1/node B.

Rubric:

- 0.5 points per correct node value.
- 1 point for decision.

b) With a brief explanation, determine the branches that can be pruned. 13 (CO1) (PO1)

Solution:

Branches 14 and 15 must be checked.
After checking branch 16, we know that $I \leq 2$. Given a choice between $H = 3$ and $I \leq 2$, the maximizer E will always prefer H, so we can stop looking at children of I. Thus *branch 17 will be pruned.*
After checking branch 18, we know that $E \geq 15$. Given a choice between $E \geq 15$ and $D = 3$, the minimizer B will always prefer D, so we can stop looking at children of E. Thus *branch 10 will be pruned.*
After checking branch 21, we know that $L \leq 2$. If the node in branch 22 is smaller or equal to 2, then it will be chosen by L. But these values will not eventually be chosen by A because we know that $B \geq 3$; If we could get larger than 2 from L, only then it will be chosen. Thus *branch 22 will be pruned.*
Then we check $M = 3$. We know that $C \leq 3$ and $B \geq 3$. Thus, the root node would prefer B over C. Thus, *branch 6 will be pruned.*

Rubric:

- 1 point for selecting each correct branch.
- 2 points for each correct explanation.
- 1 point for not choosing any wrong branch.

c) Pruning the game tree shown in Figure 2 via alpha-beta pruning may cause some problems. Identify one such problem and propose an efficient solution to it. 4 (CO3) (PO3)

Solution:

A problem that will occur is that the values of both nodes B and C will be the same.
One possible solution is to keep track of the branch (thus the corresponding action) that was first to have the highest value.

Rubric:

- 1 point for identifying the problem
- 1 point for it being specific to the given game tree.
- 1 point for the solution.
- 1 point for the solution being efficient.

d) Consider the general case where the terminal nodes of the game tree shown in Figure 2 can take on any real values instead of the given ones. Now, answer the following questions:

i. Determine the minimum and maximum number of terminal nodes that can be pruned. 6 (CO1) (PO1)

Solution:

The minimum number of nodes that can be pruned is 0 because there could always be such terminal values that requires all terminal nodes to be checked.
The maximum number of terminal nodes that can be pruned is 6. Because the nodes under the branches 14, 15, 16, 18, 21, and 23 will always have to be checked no matter what values the leaf nodes take on.

Rubric:

- 1 point for each correct value.

- 2 points for each correct explanation.

ii. Propose a new set of values for the terminal nodes so that the maximum number of those nodes can be pruned. 5 (CO3) (PO3)

Solution:

The current values result in the maximum number of pruning. Since this is a Minimax Problem, we can linearly scale all the values to get the same outcome. So we can simply increase all the values by 1 to get: $\{4, 18, 3, 13, 16, 26, 1, 3, 6, 4, 3, 15\}$.

Rubric:

- $\max(0, X - 2)$ points for pruning X nodes with the proposed values.
- 1 point explanation.